Human Activity Recognition in Smart Homes Using IMU Wristbands: A Comparative Study of Sampling Strategies and Machine Learning Models

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***Abstract*—This research explores Human Activity Recognition (HAR) for monitoring activities in smart homes using IMU wristbands, specific sampling strategies, and machine learning models to enhance accuracy. HAR systems are vital for identifying Activities of Daily Living (ADL), especially for elderly care. Data from 19 participants performing 14 activities in a smart apartment was collected using three-axis motion sensors. Preprocessing involved temporal windowing and statistical feature extraction, while dimensionality reduction was applied via PCA and SelectKBest. Four machine learning models (KNN, Random Forest, XGBoost, CatBoost) were trained with strategies like Leave-One-Subject-Out cross-validation. Longer temporal windows, such as 15 seconds, significantly improved performance. Activities like *Brushing Teeth* achieved an F1 score of 89%. While Chebyshev filtering had minimal impact, room-specific models and longer temporal windows demonstrated great potential for HAR in healthcare.**

***Index Terms*—Human Activity Recognition, IMU Wristbands, Machine Learning, Smart Homes, Elderly Care, Activities of Daily Living, Sampling Strategies**

I. Introduction

With the global population aging rapidly, healthcare systems are under increasing strain due to age-related diseases such as Alzheimer's and other cognitive impairments. The United Nations reports that life expectancy has grown significantly, with the population aged 65 and older expected to reach over 1.5 billion by 2050, doubling from 727 million in 2020 [1]. This demographic shift poses significant challenges for healthcare systems, particularly as cognitive decline can lead to difficulties in daily activities and increased dependence on caregivers. According to the World Health Organization (WHO), nearly 10% of people aged 65 and above are living with dementia, highlighting the urgent need for advanced monitoring technologies that can assist in recognizing behaviors requiring intervention and support for these individuals [2]. Research indicates that effective monitoring can enhance the quality of life for seniors, facilitating independent living while ensuring timely assistance [3], [4]. HAR systems have emerged as a solution to this growing concern, with technologies designed to track and recognize ADL. These systems play a crucial role in elderly care by providing insights into functional capacity and health status, helping caregivers intervene promptly when necessary [5]. Wearable devices, particularly those with IMUs, have gained popularity due to their ability to capture accurate real-time motion data, making them ideal for continuous monitoring in smart homes [6]. By tracking movements and analyzing physical activity, these systems enhance the quality of life for elderly individuals, facilitating independent living and reducing the burden on healthcare systems [3]. Several alternative technologies have been explored to advance HAR capabilities. Wearable sensors, such as smartwatches, accelerometers, and gyroscopes, monitor physical activities through movement tracking and physiological signals, including heart rate and body temperature [6]. Thermal cameras, which detect heat patterns to recognize ADLs, offer a non-invasive solution, but their reliance on 2D images often necessitates integration with other equipment, such as RGB cameras, for more accurate results [7]. RGB cameras, while providing detailed visual data, raise privacy concerns, particularly in settings like smart homes, making their use in elderly care limited [7]. Additionally, ambient intelligence systems, which employ noninvasive sensors like motion detectors or electrical outlet sensors, infer activities based on environmental interactions, such as the use of appliances or light switches [4]. One of the key challenges in HAR is ensuring data quality and variety for accurate classification. A major obstacle is the availability of sufficiently large and balanced datasets for

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training machine learning models. Studies have shown that the size and balance of datasets significantly impact classification performance [8], [9]. To address this, cross-validation methods such as leave-one-subject-out (LOSO) have been proposed to ensure models are generalizable across individuals with different activity patterns [7]. Moreover, in smart home environments, sensors like thermal imaging and Internet of Thing (IoT) devices have been explored for activity recognition [4]. Hence, IMU wristbands provide a unique advantage in terms of portability and accuracy while maintaining user privacy [4], [6]. This study aims to enhance Human Activity Recognition (HAR) in smart home environments by leveraging classical machine learning models and innovative data collection strategies. Unlike previous works that predominantly rely on deep learning, we take a deliberate first-step approach with interpretable and computationally efficient models. Our contributions include a participant-centered data collection methodology that ensures natural variability in movement, an evaluation of multiple sampling strategies, and an in-depth analysis of temporal windowing effects on classification performance. Given these challenges and the need for highly accurate and scalable solutions, our research proposes a comprehensive HAR framework utilizing IMU wristbands and multiple machine learning models to classify daily activities within a smart apartment environment. Specifically, we focus on recognizing activities such as *Brushing Teeth, Doing Homework, Drinking, Eating, Getting Dressed/Undressed, Going to the Toilet, Putting Away Clean Laundry, Putting Away Dishes, Reading a Book, Resting, Sleeping, Taking a Shower, Using a Computer/Smartphone, and Washing Dishes*. These activities represent common ADL in smart home environments, where accurate recognition can contribute to improved elderly care, independent living, and assistive healthcare technologies. We compare the effectiveness of various sampling strategies and machine learning techniques, aiming to improve HAR accuracy and scalability. The study achieved the highest F1-scores using a 15-second temporal window and SelectKBest feature selection, with XGBoost achieving an F1-score of 87.44% and CatBoost closely following at 87.29%. Activities such as BrushingTeeth and PuttingAwayCleanLaundry were classified with high accuracy, achieving F1-scores of 89%-92% and 65%-74%, respectively. However, activities like GoingToilet, GettingDressedUndressed, and TakingShower proved more challenging, with F1-scores ranging from 19%-57%. These findings highlight the potential of optimal configurations in enhancing classification performance across activities while addressing the complexities of HAR systems. By refining IMU-based HAR systems, our research offers a promising solution for addressing future healthcare challenges, contributing to more efficient and responsive elderly care solutions. Furthermore, to encourage further advancements in HAR, we make our dataset available upon request to researchers interested in replicating or extending this work.

II. Related work

HAR using IMUs has gained significant attention, particularly in smart home applications, where wearable wristbands track activities in real-time. Various machine learning and deep learning methods have been explored to improve accuracy and computational efficiency in healthcare, fitness, and assistive robotics [6]. Studies focus on reducing computational load for deployment on resource-limited devices. A knowledge distillation approach mapping tri-axial IMU signals to a single axis maintains high accuracy (92.90%) in HAR applications [10].

HAR has proven valuable for workplace health and safety, particularly in assistive robotics. An IMU-based LSTM deep learning system recognizes activities and estimates payloads in industrial exoskeletons, achieving a 90.80% F1-score [11]. Optimized deep learning models address class imbalance and data noise, attaining high accuracies (94.91%) aided by metaheuristic optimization [12]. Traditional machine learning models remain effective. A study comparing ANN, DTC, RFC, and KNN found RFC as the top performer (97.67%), highlighting ensemble models' strength in recognizing complex activities [12].

CNNs process IMU time-series data for fitness tracking, exceeding 94% accuracy, making them ideal for real-time processing [13]. Advances in hybrid deep learning integrate CNN and recurrent units, achieving 97.29% accuracy in gym activity classification [14]. GenerischNet, a BiGRU-CNN model, analyzes diverse human motion tasks, reaching 96.97% accuracy. Other models combining CNN and Bi-LSTM layers achieve 94.91%, reinforcing the value of optimized frameworks for handling class imbalance and noisy data [15].

Wearable IMU sensors are tailored for healthcare, with deep learning models adapted to specific sensor types to improve accuracy. A CNN-based model processing accelerometer, gyroscope, and barometer data achieved 99.35% accuracy, though potential overfitting remains a concern [16]. To enhance interpretability, Adaptive Hierarchical Decision Trees (AHDT) classify activities hierarchically, balancing complexity and interpretability in healthcare and emergency applications [17].

Cross-modal approaches improve HAR in real-world settings. IMU-based transfer learning leverages multiple sensor modalities for adaptability, improving generalization via instance- and feature-based transfer methods [10]. Hybrid models further demonstrate the benefits of combining data sources. For example, a CNN-ResBiGRU model integrates convolutional and recurrent layers to classify gym activities using multimodal IMU and EMG data, achieving high accuracy [11].

Deep learning in rehabilitation also shows promise. A framework for classifying shoulder rehabilitation exercises using IMU sensors achieved 92.5% accuracy, supporting continuous monitoring in clinical settings [13]. Unlike general-purpose models, room-specific classifiers leverage spatial context to enhance HAR in smart homes. Comparative analyses of machine learning techniques and dimensionality reduction

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methods like PCA and SelectKBest optimize HAR performance. Incorporating room data and longer temporal windows mitigates activity overlap and class imbalance, making this approach well-suited for scalable healthcare applications.

III. Methodology

A. Data Collection

The dataset was gathered from 19 participants who performed 14 different human activities within a prototype apartment designed by the Laboratory on Ambient Intelligence for Activity Recognition (LIARA). Data collection was conducted using IMU wristbands that measured motion signals along the X, Y, and Z axes at a frequency of 50 Hz. The raw data files, stored in JSON format, contained participant identifiers, activity names, and the specific rooms where the activities occurred.

Unlike many HAR studies that rely on standard datasets, our research innovates in data collection by designing an authentic, participant-centered approach. The data was collected in a realistic setting, allowing participants to perform activities freely, rather than following pre-defined, constrained sequences. This methodology enhances the natural variability of movement patterns, making the dataset closer to real-life conditions and improving its applicability in practical smart home environments. The dataset is available upon request from the authors of this study.

B. Data Preprocessing

To ensure the highest quality of data, several preprocessing steps were performed. First, temporal alignment was conducted to synchronize the data, eliminating irrelevant sections such as delays in the start and stop of activities by cropping the first 10 seconds and the last 5 seconds of each recording. Chebyshev filtering was employed to enhance signal quality by reducing high-frequency noise, which could obscure activity-specific patterns. The data was then divided into time windows of 5, 10, and 15 seconds. In this phase, various features were extracted from the motion signals to provide a comprehensive representation of the data. The extracted features included: Minimum (min), Maximum (max), Mean (average), Median (median), Standard Deviation (std), Skewness (skewness), Kurtosis (kurtosis), Mean Absolute Deviation (MAD), Waveform Length (WL), Zero-Crossing Rate (ZCR), Mean Crossing Rate (MCR), Squared Energy (Energies), PSD Entropy (PSD Entropy), and 3 levels of wavelet energy (cA3, cD3, cD2, cD1). Each instance in the dataset contained 42 attributes, capturing a comprehensive range of statistical and temporal features for effective activity classification. PCA and SelectKBest were utilized for feature selection. PCA reduced features to 7–8 principal components, achieving a balance between dimensionality reduction and retaining relevant data variance. SelectKBest, using statistical significance tests, consistently selected the top N features, maximizing classification accuracy.

C. Model Training

Rather than relying on deep learning approaches, this research takes a deliberate step toward exploring classical machine learning methods as an initial approach to understanding HAR in smart homes. The choice to exclude deep learning models at this stage is intentional, allowing a focus on interpretability, computational efficiency, and adaptability. By prioritizing classical machine learning and data processing techniques, we aim to establish a strong baseline, ensuring a clear evaluation of fundamental methodologies before progressing to more complex architectures.

We implemented and evaluated four machine learning models: K-Nearest Neighbors (KNN), Random Forest, XGBoost, and CatBoost [4].

We implemented and evaluated four machine learning models: K-Nearest Neighbors (KNN), Random Forest, XGBoost, and CatBoost [4]. The generalization of each model was tested by training it using different sampling strategies. For this project, three sampling methods were developed to split instances into training, validation, and testing sets (Train-Validation-Test). The validation set plays a crucial role in classical machine learning as it is used for hyperparameter optimization, ensuring that the model is fine-tuned effectively without overfitting to the training data. **Strategy 1** (S1): 70%-15%-15% of the activity files files were aggregated and randomly split based on these percentages for deployment. **Strategy 2** (S2): For an activity-specific list of participants participated, the list of participants was shuffled and split into 70%-15%-15% participant-specific data. **Strategy 3** (S3): Leave-one-subject-out strategy that for each iteration, selects one of the participants as test set, picks one randomly as validation set and uses the rest for training. We have 19 participants in the study, resulting in 19 models (one per participant as the test set) using this approach. These sampling methods ensure comprehensive evaluation of the models across various scenarios. Additionally, LOSO cross-validation was employed to further assess the models' generalization capabilities in real-world settings [7].

KNN: The KNN model is a straightforward yet effective machine learning algorithm that classifies instances based on their proximity to training samples in the feature space. When a new instance is introduced, KNN computes the distance between the new instance and all training instances. It then identifies the k nearest neighbors and predicts the class based on majority voting. KNN is particularly useful for its simplicity and ease of interpretation, making it a popular choice for various applications, including healthcare [6]. **Random Forest**: This model builds an ensemble of decision trees to improve classification performance. Each tree is constructed using a random subset of the data's features, ensuring that the model does not overfit to any particular dataset. During training, 70% of the features were randomly selected at each node, and the maximum depth of the trees was optimized to avoid overfitting. Random Forest's ability to handle large datasets with high dimensionality makes it an ideal candidate for HAR

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tasks, where multiple features are extracted from sensor data [18]. **XGBoost**: The Extreme Gradient Boosting (XGBoost) model is a powerful ensemble algorithm that utilizes multiple decision trees to produce a final prediction. Unlike Random Forest, where all trees contribute equally, XGBoost assigns different weights to each tree based on their contribution to the prediction error, employing gradient descent to minimize the loss function iteratively [19]. This model is highly efficient and often used in competitive machine learning due to its speed and performance, particularly in scenarios with complex data distributions [18]. **CatBoost**: CatBoost operates similarly to XGBoost but has the significant advantage of natively handling categorical variables, eliminating the need for preprocessing steps like encoding. This capability simplifies data preparation and improves model performance [20]. CatBoost is designed to work well with various types of data and is particularly effective for datasets with a mix of categorical and numerical features, making it a suitable choice for real-world applications.

D. Evaluation

The aim of this section is to evaluate the performance of several machine learning algorithms for human activity recognition using IMU sensors, with preprocessing steps involving dimensionality reduction using Principal Component Analysis (PCA) or feature selection via the SelectKBest algorithm.

a) Dimensionality Reduction and Feature Selection: To avoid overfitting of machine learning models, dimensionality reduction was applied using PCA, with a cumulative variance threshold set at 95%. This threshold helps limit the number of principal components while retaining a large proportion of the original data's information. Also, we tested SelectKBest for feature selection, using a score of 1000 applied to the **f_classif** metric to determine the most relevant features. This threshold was set through trial and error to optimize model performance and eliminate less impactful features. Features scoring below this threshold were excluded from training, ensuring a more efficient and streamlined selection process. This approach ensures more efficient models by focusing on the most discriminative features, reducing redundant information and to avoid overfitting.

b) Performance Measurement per Iteration: The evaluation was conducted by progressively increasing the number of features or PCA vectors used in different training iterations. For example, if 16 features were selected using SelectKBest, each model was trained successively with 1, 3, 5, 7, 9, 11, 13, and 15 features. This process resulted in eight performance measurements, each corresponding to a specific number of features, allowing for an observation of how model performance evolved as the dataset complexity increased.

c) Hyperparameter Optimization and Cross-Validation: Hyperparameter optimization and cross-validation were critical to ensuring robust model performance and generalization to new participants. Fine-tuning of the hyperparameters for the different models was conducted using the Hyperopt library, leveraging the Tree of Parzen Estimators (TPE) method. This Bayesian optimization technique efficiently selects hyperparameters to test based on previous performance results, reducing the search space and increasing iteration efficiency. Unlike traditional methods such as Grid Search, which exhaustively tests all parameter combinations, or Random Search, which randomly selects combinations, Hyperopt uses probabilistic modeling to focus on the most promising regions of the parameter space, significantly reducing computation time while improving model performance. For each machine learning model, a specific search space was defined for the hyperparameters, such as the number of neighbors for KNN models, the number of estimators for Random Forests, or the learning rate for boosting models like XGBoost and CatBoost. Additionally, cross-validation was performed using LOSO method to ensure the models generalized well to new participants. In this approach, each participant's data was excluded from the training set during one iteration and used solely for testing, mimicking real-world scenarios of applying the model to unseen individuals [21]. This rigorous combination of cross-validation and efficient hyperparameter tuning allowed for optimal configurations, such as tuning the number of decision trees, learning rates, and maximum tree depths for models like Random Forest and XGBoost, leading to better performance with reduced computational overhead [22], [23].

d) Model Evaluation: Model performance was assessed using multiple classification metrics, including accuracy score, F1 score, and the kappa coefficient, for the training, validation, and test sets of the different models. Additionally, classification scores for each activity were recorded in a text file, allowing for a comparison of the models using a confusion matrix. Furthermore, instance reports were generated to determine the presence of different participants, the room where classification occurred, and the activity being performed. The goal of these reports was to calculate the classification error percentage for each instance to analyze whether factors such as room had an impact on the classification results. Finally, several performance metrics graphs were generated to show how the models' performance evolved as the number of principal component vectors or selected features increased.

IV. Results and discussion

A. S1: 70%-%15-%15 Split per activity

Regarding the strategy of sampling S1, there are many important differences of performances between each temporal window. The obtained results show that the best performance is occurred when the temporal window is longer compared to short window. Considering that we record a higher quality of movement on a time window of 15 sec vs 5 sec, the ML Algorithms have a better capacity when classifying each activity of learning in a longer temporal space.

1) Class1 : A model for the entire apartment:

a) With PCA: On a 5 sec temporal window with dimensionality reduction PCA, we obtain models that perform on a level of 52% to 55% of accuracy score without using the Chebyshev filter.

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A performance plateau is observed starting from the 7th or 8th principal component, where the explained variance reaches approximately 90%. For the F1-Score metric, we obtain results that are roughly equivalent to the accuracy metric, around 55% depending on the learning model without filtering of the data and 50% with filtering.

The dataset shows a consistent distribution of activities, avoiding statistical bias due to class imbalance. Activities like *BrushingTeeth* (75%), *DoingHousework* (69%), and *Eating* (63%) perform best across models, while *GettingDressedUndressed* (28%), *GoingToilet* (34%), and *TakingShower* (42%) are harder to classify.

However, when examining the classification reports of the models, there is not a significant difference between the precision and recall scores across the various activities, as shown in Table I bellow.

TABLE I
F1-SCORE BY MACHINE LEARNING ALGORITHM ON A 5-SECOND TIME WINDOW, WITHOUT FILTERING AND WITH PCA

Activity	***KNN***	***CatBoost***	***Random Forest***	***XGBoost***
BrushingTeeth	77%	75%	79%	79%
DoingHousework	60%	69%	67%	70%
Drinking	59%	64%	58%	59%
Eating	63%	63%	64%	63%
GettingDressedUndressed	30%	28%	27%	29%
GoingToilet	33%	34%	38%	37%
PuttingAwayCleanLaundry	54%	58%	58%	58%
PuttingAwayDishes	57%	59%	60%	59%
ReadingBook	44%	53%	54%	51%
Resting	49%	50%	45%	52%
Sleeping	60%	62%	62%	62%
TakingShower	39%	42%	46%	45%
UsingCompSPhone	52%	56%	55%	60%
WashingDishes	53%	56%	54%	53%
Average	52.14%	54.93%	54.79%	55.50%
Standard Deviation	-	12.96%	12.86%	12.73%

In addition, the confusion matrix (obtained from XGBoost model) in Fig. 1 reveals frequent misclassification of similar activities, such as *Eating* as *Drinking* (143 instances), *Resting* as *Sleeping*, and *UsingCompSPhone* as *ReadingBook*. These challenges persist even with longer time windows and Chebyshev filtering.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1 - BrushingTeeth	1050	13	9	8	8	24	26	49	0	0	0	7	0	122
2 - DoingHousework	28	648	5	9	52	60	84	60	0	3	4	35	0	7
3 - Drinking	18	0	808	143	0	82	3	22	103	49	36	0	49	28
4 - Eating	38	3	85	971	0	1	11	152	86	105	3	0	32	5
5 - GettingDressedUndressed	38	41	26	46	239	98	140	62	29	2	22	221	0	33
6 - GoingToilet	9	16	46	13	72	509	45	43	120	33	123	36	108	147
7 - PuttingAwayCleanLaundry	24	55	0	36	124	56	901	121	0	0	62	84	0	39
8 - PuttingAwayDishes	3	30	51	32	33	17	113	986	13	0	0	4	3	54
9 - ReadingBook	0	0	103	40	0	29	0	3	787	200	5	0	272	71
10 - Resting	6	0	60	52	14	148	0	1	10	624	322	8	72	27
11 - Sleeping	21	17	22	13	37	24	38	13	16	261	946	79	2	18
12 - TakingShower	51	96	20	46	152	158	195	70	0	0	19	579	0	58
13 - UsingCompSPhone	26	0	155	55	0	80	9	0	225	173	9	0	779	0
14 - WashingDishes	27	16	35	88	35	56	34	381	16	0	6	4	20	784

Fig. 1. Confusion matrix for XGBoost with 22 components on a 5-second time window without filtering.

b) With SelectKBest: With SelectKBest, larger time windows improve performance, with Chebyshev filtering benefiting 15-second windows as shown in Table II .

Confusion matrices highlight issues in distinguishing similar activities. For example, *Sleeping* vs. *Resting* and *ReadingBook* vs. *UsingCompSPhone* are often misclassified due to minimal movement. However, distinct activities like *BrushingTeeth* achieve high accuracy (88.3% of instances correctly classified) which is clear in Fig. 2.

TABLE II
F1-SCORE COMPARISON ACROSS MACHINE LEARNING MODELS

Activity	***KNN***	***RF***	***XGB***	***CB***
BrushingTeeth	89%	89%	89%	88%
DoingHousework	63%	54%	58%	58%
Drinking	66%	72%	69%	67%
Eating	62%	71%	72%	70%
GettingDressedUndressed	42%	34%	35%	40%
GoingToilet	19%	32%	33%	26%
PuttingAwayCleanLaundry	65%	71%	74%	73%
PuttingAwayDishes	59%	70%	70%	69%
ReadingBook	60%	54%	47%	43%
Resting	45%	65%	61%	66%
Sleeping	48%	46%	48%	55%
TakingShower	45%	55%	57%	57%
UsingCompSPhone	63%	59%	57%	51%
WashingDishes	61%	71%	70%	68%
Average	56.21%	60.21%	60.00%	59.36%
Std. Dev.	15.41%	15.18%	15.00%	15.22%

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1 - BrushingTeeth	324	4	0	3	0	1	4	13	0	0	0	1	0	17
2 - DoingHousework	0	147	1	4	8	56	13	16	0	0	1	23	4	6
3 - Drinking	0	0	247	75	0	4	0	0	36	5	7	4	0	0
4 - Eating	0	1	26	363	0	0	0	0	41	0	3	0	3	17
5 - GettingDressedUndressed	17	8	0	0	82	17	60	6	0	0	0	53	0	22
6 - GoingToilet	2	1	5	56	32	107	2	7	24	7	8	12	58	29
7 - PuttingAwayCleanLaundry	0	22	0	0	50	3	334	17	0	0	1	23	0	2
8 - PuttingAwayDishes	6	0	0	7	1	0	25	274	0	0	0	1	0	64
9 - ReadingBook	0	0	0	16	0	25	0	0	271	35	70	0	34	4
110 - Resting	0	0	6	11	1	20	0	0	15	289	31	3	0	0
11 - Sleeping	0	7	11	3	12	50	4	5	1	167	173	21	0	0
12 - TakingShower	0	71	0	1	25	4	37	6	0	0	1	180	0	3
13 - UsingCompSPhone	8	0	12	20	0	3	0	0	161	13	5	0	233	0
14 - WashingDishes	3	4	0	8	4	23	9	58	0	0	2	3	0	344

Fig. 2. Confusion matrix for the random Forest Model on a 15-Second time window with Chebyshev filter

2) Class 3 A room-specific model within the apartment: Creating room-specific models within the apartment (Class3) improves machine learning performance by leveraging the fact that certain activities only occur in specific rooms. This reduces classification ambiguity by eliminating impossible activities.

a) Bathroom: The classification models for the bathroom focus on three distinct activities: *BrushingTeeth*, *GoingToilet*, and *TakingShower*. PCA-based models achieve accuracies between 76% and 87%, with better performance for larger time windows, though the Chebyshev filter shows no significant impact and can sometimes decrease performance. However, models using SelectKBest feature selection outperform PCA, achieving accuracies of 81% to 88%, especially with smaller time windows. For a 15-second window, results between PCA and SelectKBest are comparable, and performance across machine learning algorithms is similar. Notably, detecting *GoingToile* remains challenging for all models, with high recall rates indicating frequent misclassifications. Given these findings, we will present only the results using SelectKBest without the filter to emphasize the improved performance and

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simplicity of interpretation and we demonstrate the results bellow in tableIII.

TABLE III
CLASSIFICATION REPORT OF DIFFERENT MACHINE LEARNING ALGORITHMS WITH A FILTER AND A 15-SECOND TIME WINDOW FOR 29 COMPONENTS.

Activity	*Precision*	*Recall*	*F1-Score*	*Support*
KNN				
BrushingTeeth	91%	94%	92%	362
GoingToilet	85%	82%	84%	370
TakingShower	86%	86%	86%	331
Average	87%	87%	87%	-
Random Forest				
BrushingTeeth	92%	94%	93%	362
GoingToilet	87%	81%	84%	370
TakingShower	85%	89%	87%	331
Average	88%	88%	88%	-
CatBoost				
BrushingTeeth	91%	95%	92%	362
GoingToilet	86%	82%	85%	370
TakingShower	89%	86%	88%	331
Average	88%	88%	88%	-
XGBoost				
BrushingTeeth	88%	94%	91%	362
GoingToilet	85%	81%	83%	370
TakingShower	85%	84%	85%	331
Average	86%	86%	86%	-

Machine learning models perform poorly in the bedroom compared to the entire apartment, with accuracy and F1-scores ranging from 47% to 68%. Longer time windows improve performance, but both PCA and SelectKBest face challenges due to imbalanced activity distributions, leading to biases. Some activities, like *UsingCompSPhone*, show large variations between models, while others, like *Resting* and *Sleeping*, are frequently confused. Distinctive activities, such as *Drinking*, perform better, but subtle activities, like *DoingHousework*, struggle with low recall despite high precision as shown in figure Fig. 3.

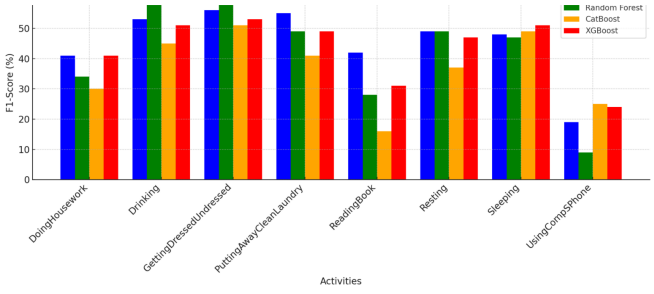


Fig. 3. Comprehensive performance analysis of machine learning models for activities in the bedroom

b) Dining Room: The performance inconsistencies for the dining room stem from the methodology used to create training, validation, and test sets. Due to insufficient instances of the activity *Drinking*, no data was included in the validation set, causing machine learning models to struggle with recognizing this activity, leading to artificially low F1-scores. However, the remaining activities performed well, with models achieving over 80% accuracy in most cases when using PCA or SelectKBest, even without the Chebyshev filter. Performance improved with longer time windows, and high Cohen-Kappa scores indicate overall model reliability. Despite imbalanced activity representation, there is no statistical bias, as other activities are correctly classified, emphasizing the distinctiveness of movements in this room and we can notice that in Fig. 4.

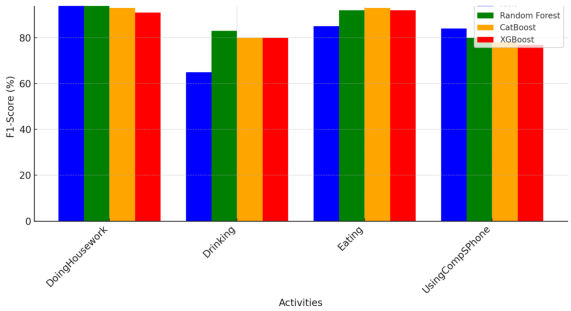


Fig. 4. F1-Score for dining room activities using a 15-second window and SelectKBest with a Chebyshev filter

c) Kitchen: For the kitchen, machine learning models performed well with PCA, achieving F1-scores and Cohen-Kappa values around 93% for a 15-second time window. However, SelectKBest yielded lower performance due to the importance threshold of 1500, which limited the number of features to just 2–8 in many scenarios, preventing models from reaching their performance plateau. Despite this, classification results showed no statistical bias, with balanced activity classes and consistently high precision, recall, and F1-scores across models and activities. Notably, while the models struggled with activities like *DoingHousework*, they successfully distinguished between similar activities, such as *PuttingAwayDishes* and *WashingDishes*, which were often confused in earlier apartment-wide models, and the results are demonstrated in Fig. 5.

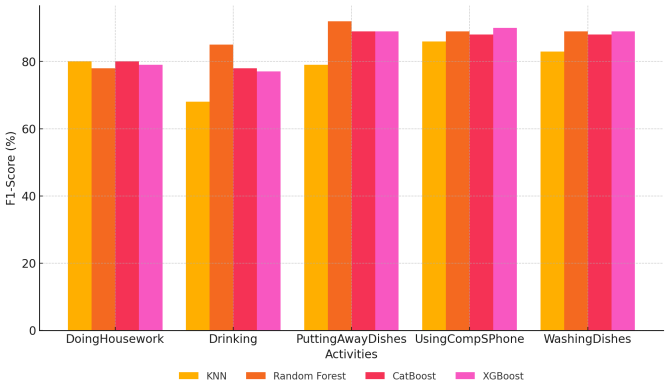


Fig. 5. F1-Score for kitchen activities using a 15-second window and PCA with a Chebyshev filter

d) Living Room: For the living room, the performance of machine learning models was limited by the thresholds set for SelectKBest and PCA, which restricted the number of features

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used, particularly for longer time windows. As a result, key metrics, such as F1-scores, decreased for these windows, with the Chebyshev filter on a 5-second window delivering the best results. Despite generally high performance, some activities, such as UsingCompSPhone, were not recognized by Random Forest or CatBoost, while kNN and XGBoost handled them effectively. Overall, activity-level performance hovered around 50% for precision and recall, except for a few well-performing cases. A larger feature set for a 15-second window could potentially improve these results further, as we can see in Fig. 6.

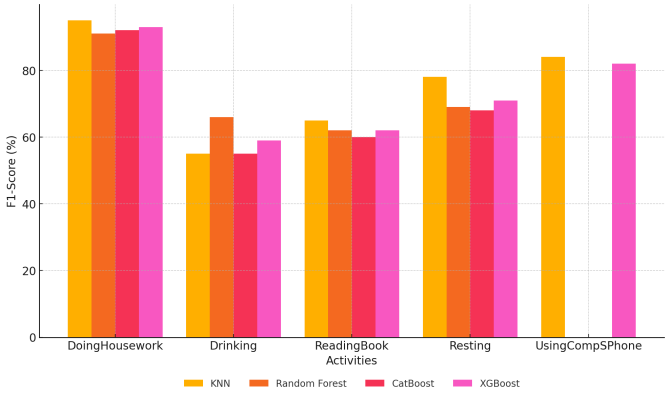


Fig. 6. F1-Score for living room activities using a 15-second window and SelectKBest with a Chebyshev filter

B. S2 : 70%-15%-15% Sampling Strategy per Participant

The S2 sampling strategy by participant shows classification performance similar to the S1 strategy, indicating that the sampling method does not significantly improve human activity classification. Key observations remain consistent: longer time windows improve model performance, the Chebyshev filter has minimal impact, and static activities are more challenging for machine learning models to recognize.

1) Class1:

a) With PCA: Activities with minimal recorded movement are the hardest to predict, but using a model based on the entire apartment balances the training, validation, and test sets, improving model training. The poorly performing activities in S2 remain consistent with S1, showing that the sampling strategy does not enhance classification performance and it is clear in Fig. 7.

b) With SelectKBest: Comparing feature selection with SelectKBest and dimensionality reduction with PCA reveals no significant performance improvement between the two approaches. Overall accuracy ranges between 50% and 60% depending on the time window, indicating PCA is not necessary for enhancing model performance. Both methods struggle to classify activities with minimal movement, showing similar results like shown in Fig. 8.

c) Impact of Including Room as a Feature in Human Activity Classification Models: Including the room as a feature in the classification model significantly improves performance,

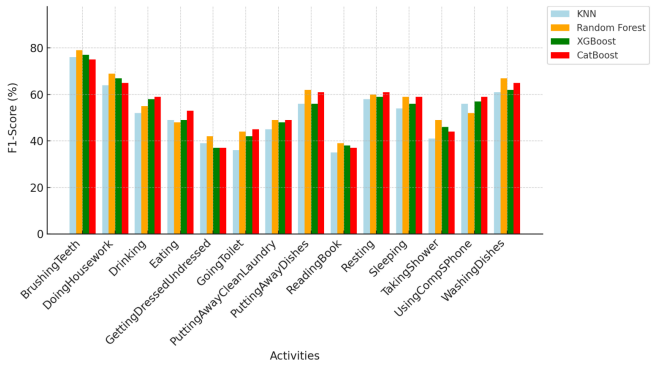


Fig. 7. F1-Score by machine learning algorithms using a 15-second time window, without filtering, and with PCA

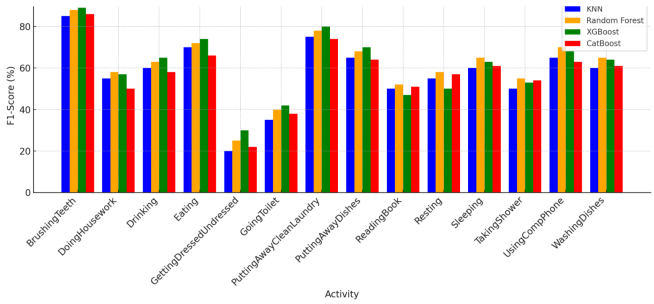


Fig. 8. F1-Score by machine learning algorithms using a 15-second time window, without filtering, and with SelectKBest

with overall accuracy increasing to 73%-78% and the lowest F1-score rising from 20% to 49%. The room emerges as the most important variable for classifying human activities, highlighting its value in reducing misclassifications. Further improvements could involve merging similar activities like *Resting* and *Sleeping* and analyzing the confusion matrix for additional insights that are represented in Fig. 9.

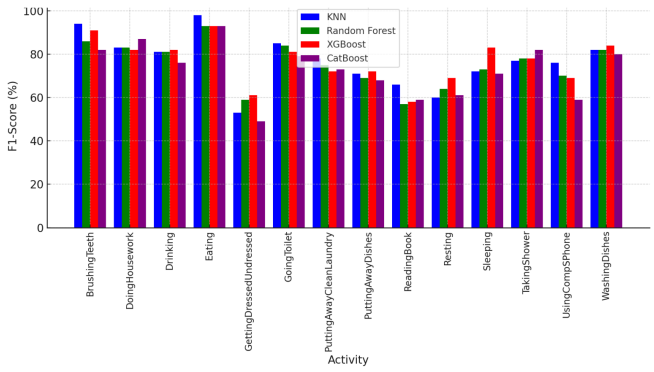


Fig. 9. F1-Scores for various activities by machine learning algorithms over a 15-second time window, without filtering, and using SelectKBest, with the dataset incorporating room as an attribute

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V. Conclusion

In this study, we conducted a comprehensive evaluation of HAR using IMU wristbands in a smart home environment. By employing advanced machine learning models and various sampling strategies, we analyzed the impact of preprocessing steps, dimensionality reduction, and temporal window sizes on classification performance. This work stands out for its unique combination of real-world experimental setups with 19 participants, diverse activities, and multiple models, providing valuable insights into HAR systems for healthcare applications.

Our results demonstrate the effectiveness of longer temporal windows in improving classification accuracy, particularly for complex activities. The inclusion of room-specific features significantly enhanced model performance by reducing activity overlap. However, preprocessing techniques like Chebyshev filtering and PCA showed only marginal improvements, highlighting the need for context-specific preprocessing strategies. The findings underscore the potential of HAR systems in elderly care by providing reliable and scalable solutions for real-world scenarios.

For future work, we propose exploring deep learning architectures, such as hybrid models combining CNNs and RNNs, to capture both spatial and temporal features more effectively. Additionally, integrating contextual data, such as environmental conditions or behavioral patterns, could further improve the system's adaptability. Addressing challenges like class imbalance and developing lightweight models for deployment on resource-constrained devices will also be critical in advancing HAR systems for practical use.

References

[1] U. Nations, "World population ageing 2019: Highlights," 2019, population Division, Dept. Econ. Social Affairs, United Nations, New York, NY, USA, p. 46.

[2] W. H. Organization, "Falls: Key facts," 2021, available: https://www.who.int/news-room/fact-sheets/detail/falls.

[3] D. Beltz, A. Patel, and K. Reid, "Adl/iadl and quality of life in elderly populations: A longitudinal study," *J. Gerontology*, vol. 67, no. 5, pp. 453–465, 2022.

[4] F. Shah, C. Kalra, and S. J. Vassallo, "Human activity recognition using ambient intelligence and iot: A survey," *IEEE Internet Things J.*, vol. 7, no. 5, pp. 4533–4547, 2020.

[5] R.-L. Heikkinen, "Who ageing and health programme report," 2010, world Health Organization, Available: https://apps.who.int/iris/handle/10665/65230.

[6] A. Avci, H. Lach, M. Buchner, and B. Yang, "Activity recognition using inertial sensing for healthcare, well-being and sports applications: A survey," *IEEE Trans. Instrumentation Meas.*, vol. 59, no. 11, pp. 437–444, 2010.

[7] M. Gholamiangonabadi, M. Avots, and A. Prange, "Leave-one-subject-out cross-validation in human activity recognition for elderly care: A critical assessment," *J. Ambient Intell. Humanized Comput.*, vol. 11, no. 6, pp. 2339–2350, 2020.

[8] J. Althnian, A. AlSkaif, and M. Fiore, "Impact of dataset size on classification performance: A comparative study in human activity recognition," *Sensors*, vol. 21, no. 12, pp. 3112–3124, 2021.

[9] Y. Chen, G. Mao, and L. Xu, "Impact of data volume and dataset balance in deep learning for human activity recognition," *IEEE Access*, vol. 6, pp. 35 053–35 060, 2017.

[10] Y. Zhang and P. Liu, "A survey of imu-based cross-modal transfer learning in human activity recognition," *IEEE Trans. Artif. Intell.*, vol. 5, no. 2, pp. 128–136, 2024.

[11] L. Chen and M. Zhao, "A residual deep learning method for accurate and efficient recognition of gym exercise activities using electromyography and imu sensors," *IEEE Sensors J.*, vol. 23, no. 5, pp. 1972–1981, 2023.

[12] J. C. S. Lee and K. Kim, "Generisch-net: A generic deep model for analyzing human motion with wearable sensors in the internet of health things," *IEEE Internet Things J.*, vol. 10, no. 3, pp. 2155–2168, 2024.

[13] J. P. et al., "Deep learning model for classifying shoulder pain rehabilitation exercises using imu sensor," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 4, pp. 2990–2998, 2023.

[14] M. Singh and A. Gupta, "Wearable approaches for non-invasive monitoring of tendons: A scoping review," *IEEE Rev. Biomed. Eng.*, vol. 17, pp. 203–215, 2024.

[15] C. Wu and Z. Liu, "Non-stationary bert: Exploring augmented imu data for robust human activity recognition," *IEEE Trans. Mobile Comput.*, vol. 22, no. 7, pp. 1823–1835, 2024.

[16] M. B. et al., "Visualizing the unseen: Illustrating and documenting phantom limb sensations and phantom limb pain with cala," *Front. Rehabil. Sci.*, vol. 3, p. 806114, 2022.

[17] S. P. R. Thomas and J. Martin, "Otago exercises monitoring for older adults by a single imu and hierarchical machine learning models," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 145–156, 2023.

[18] D. Martin and A. Roy, "An optimized deep learning model for human activity recognition using inertial measurement units," *Pattern Recognition Letters*, vol. 151, pp. 45–53, 2023.

[19] R. L. Doran, M. V. Kossinets, and K. J. McDade, "Classification performance of supervised learning methods in physical activity detection," *J. Sports Analytics*, 2023.

[20] S. Patel and M. Singh, "Human activity recognition using machine learning algorithms based on imu data," *Sensors and Actuators A*, vol. 276, pp. 65–78, 2023.

[21] R. G. et al., "Ensemble deep learning network for enhancing performances of sensor-based physical activity recognition based on imu sensor data," *IEEE Internet Things J.*, vol. 9, no. 10, pp. 10 552–10 560, 2023.

[22] H. Y. F. Xu and Y. Liu, "Human activity recognition via wi-fi and inertial sensors with machine learning," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 1251–1263, 2023.

[23] K. C. et al., "Accuracy comparison of cnn, lstm, and transformer for activity recognition using imu and visual markers," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 4, pp. 885–899, 2023.

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